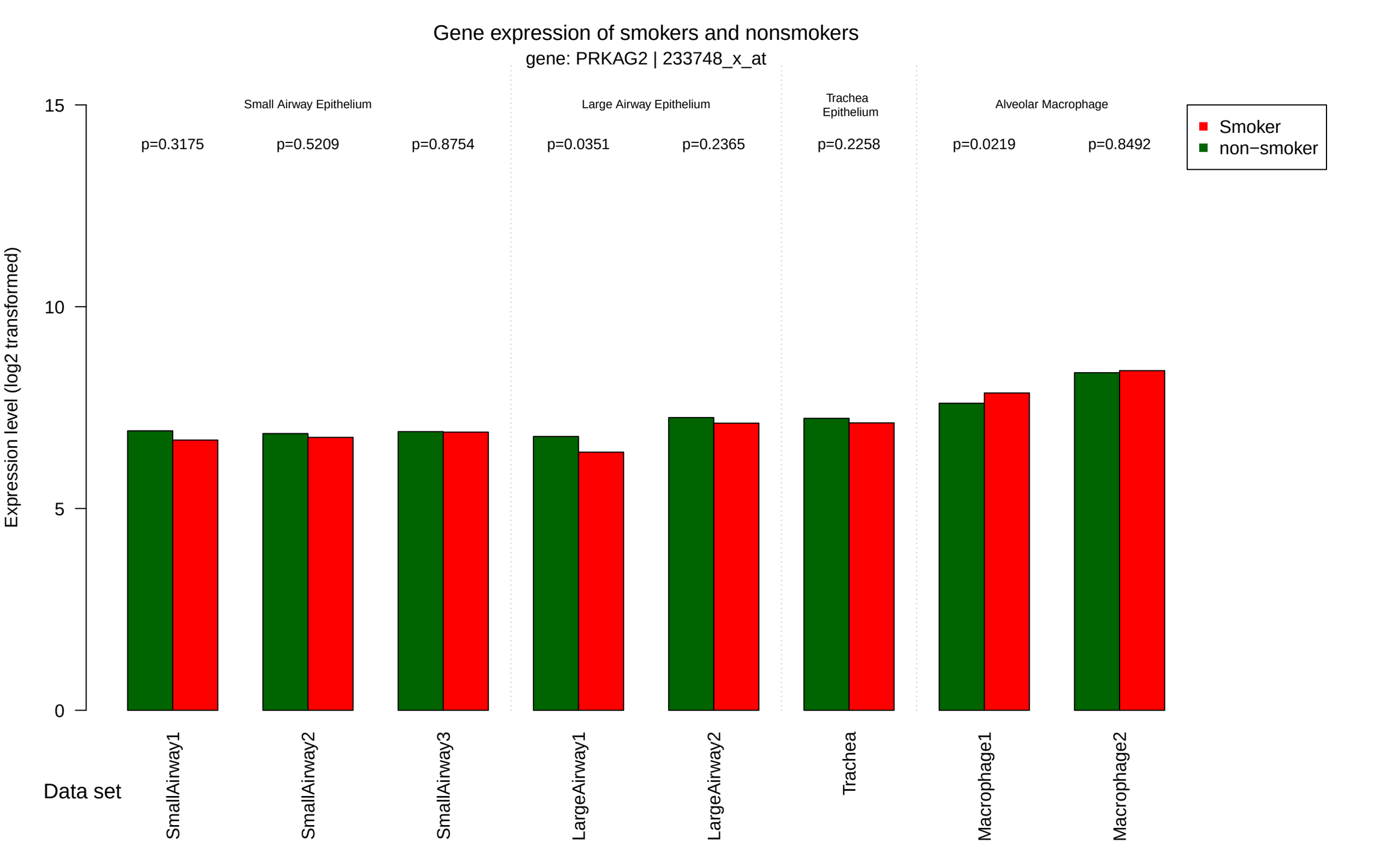
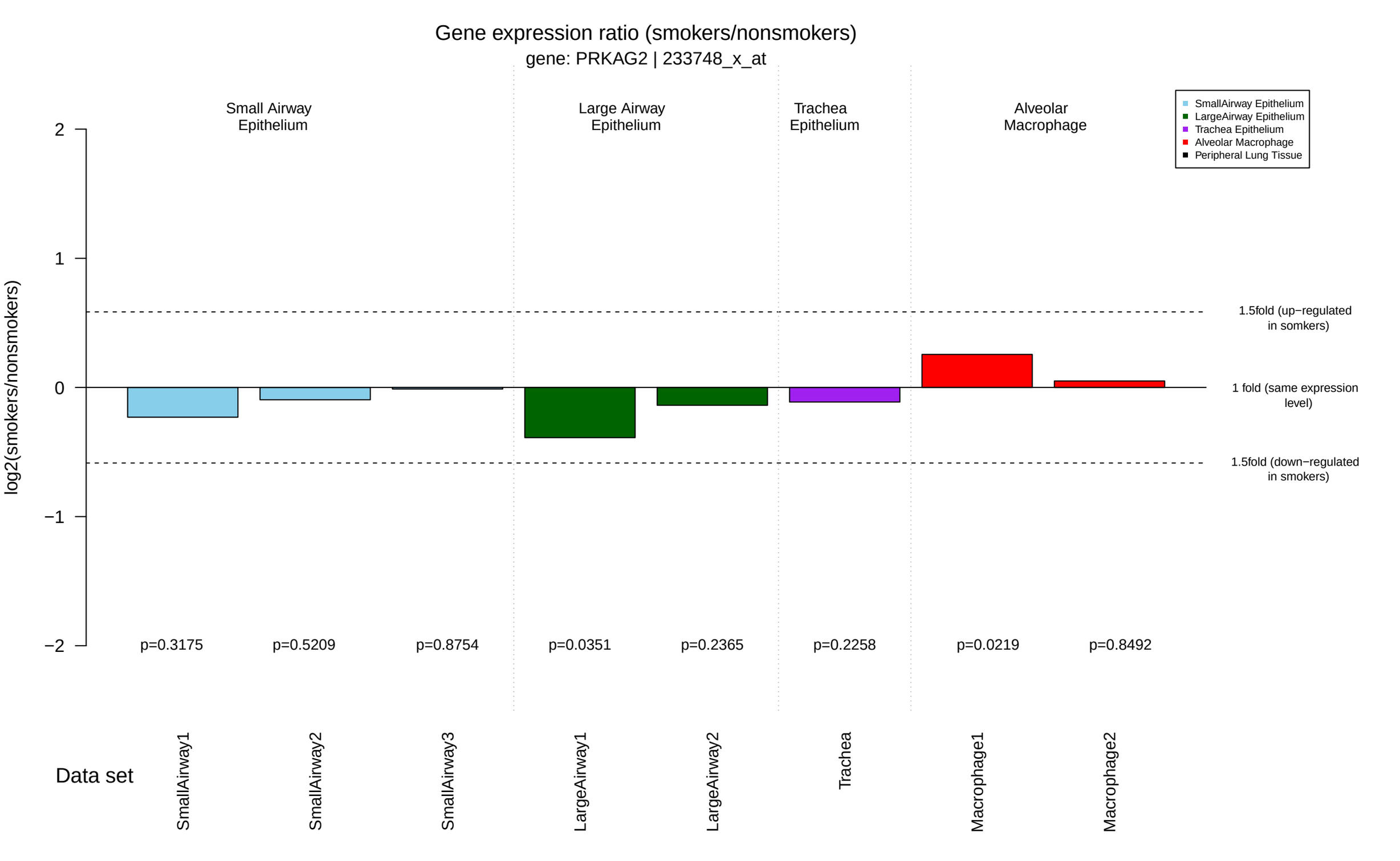
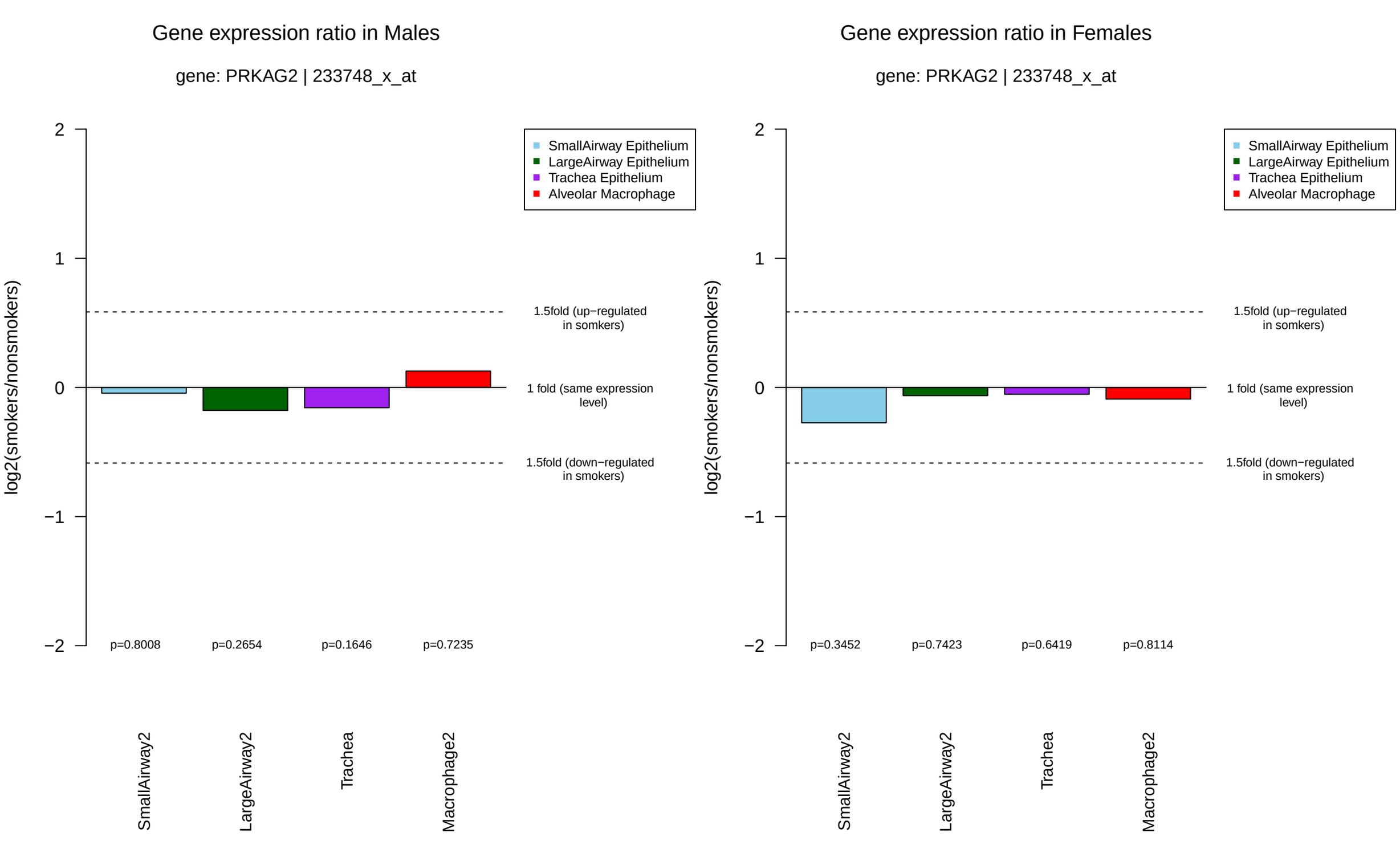
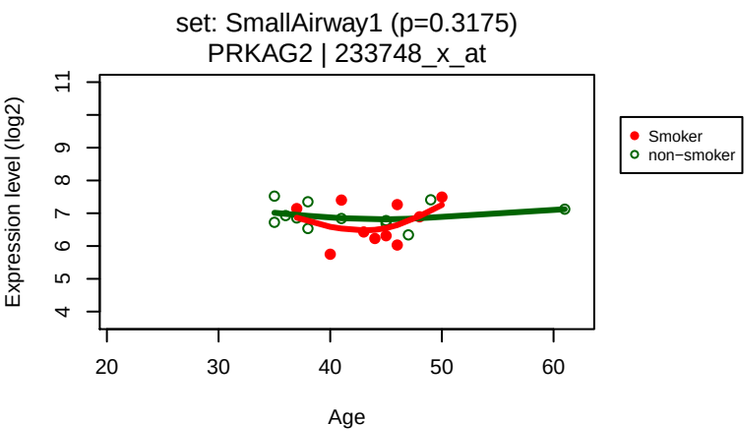
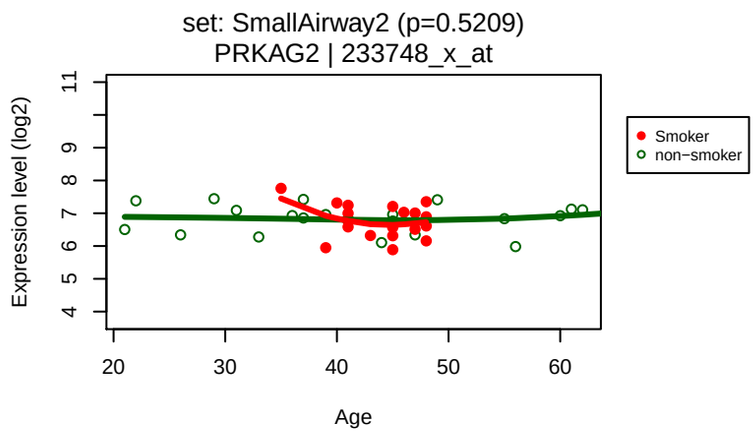
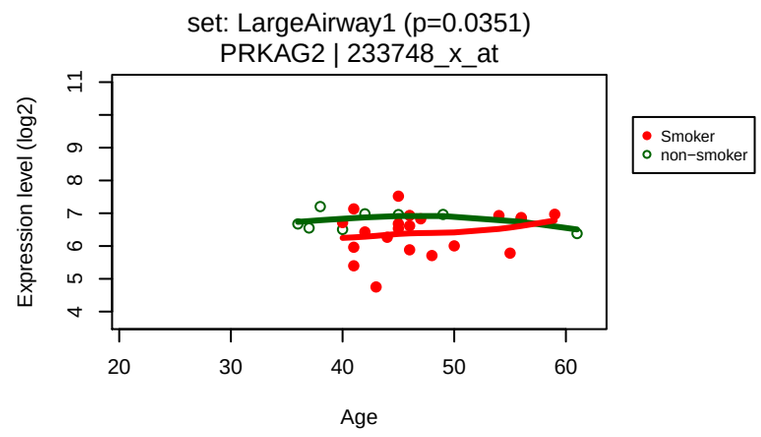

Gene expression of smokers and nonsmokers
gene: PRKAG2 | 233748_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3175
p=0.5209
p=0.8754
p=0.0351
p=0.2365
p=0.2258
p=0.0219
p=0.8492
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

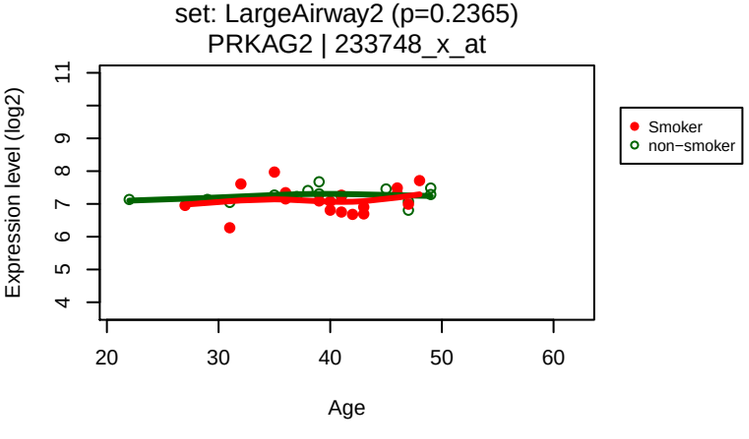

Gene expression ratio (smokers/nonsmokers)
gene: PRKAG2 | 233748_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3175
p=0.5209
p=0.8754
p=0.0351
p=0.2365
p=0.2258
p=0.0219
p=0.8492
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

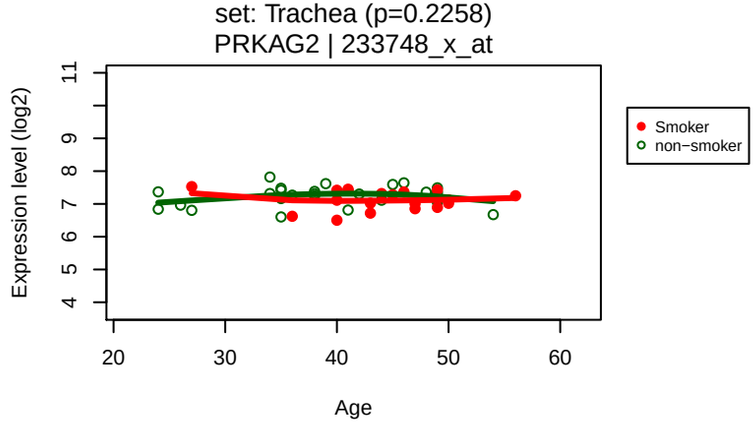

Gene expression ratio in Males
gene: PRKAG2 | 233748_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8008
p=0.2654
p=0.1646
p=0.7235
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAG2 | 233748_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3452
p=0.7423
p=0.6419
p=0.8114
SmallAirway2
LargeAirway2
Trachea
Macrophage2

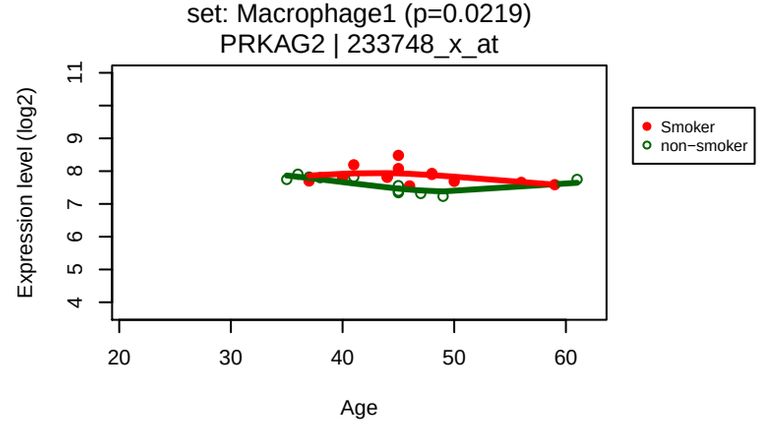

set: SmallAirway1 (p=0.3175)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker

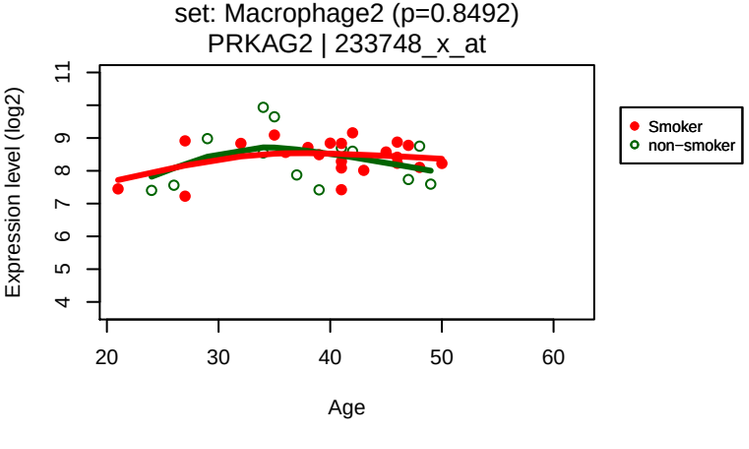

set: SmallAirway2 (p=0.5209)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker

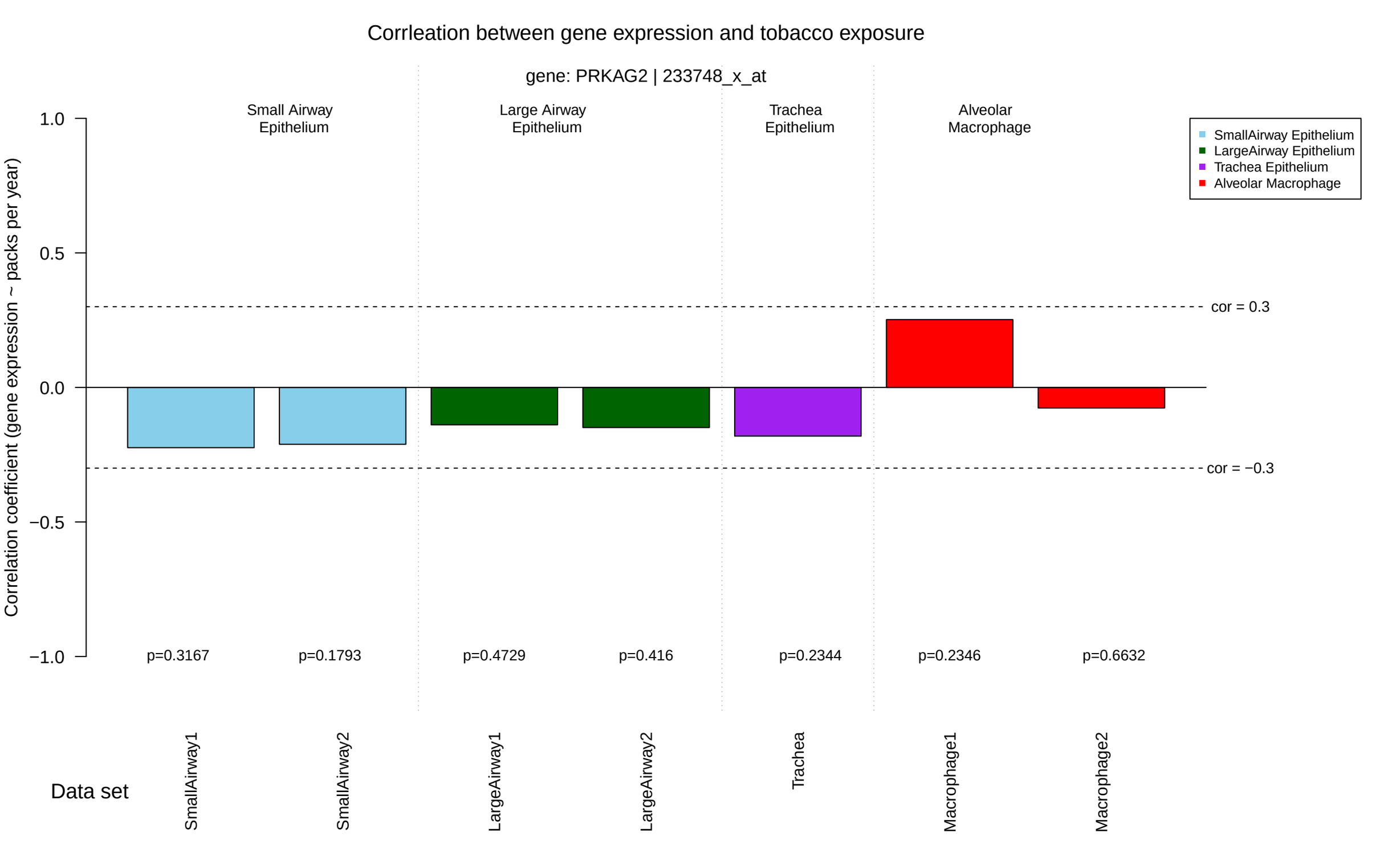

set: LargeAirway1 (p=0.0351)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2365)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2258)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0219)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8492)
PRKAG2 | 233748_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAG2 | 233748_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3167
p=0.1793
p=0.4729
p=0.416
p=0.2344
p=0.2346
p=0.6632
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2